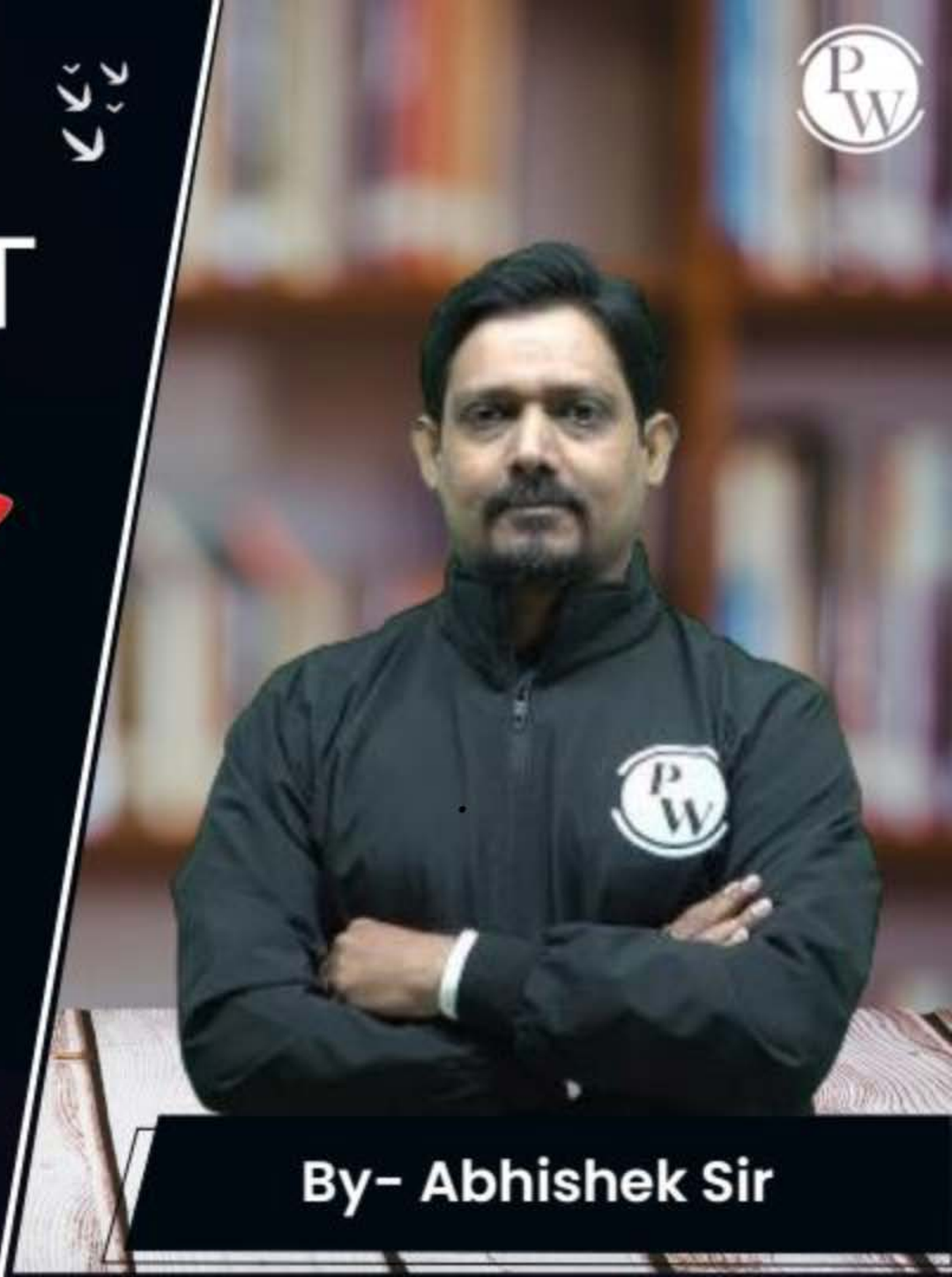
Computer Science & IT



Fundamental of C Language

Function & Storage Class

Lecture No. 02



By- Abhishek Sir

Recap of Previous Lecture



- Topic — function
- Topic — forward declaration & definition
- Topic — Control transfer control returns
- Topic — Call by value
- Topic — Actual parameter & formal parameter

Slide

Topics to be Covered

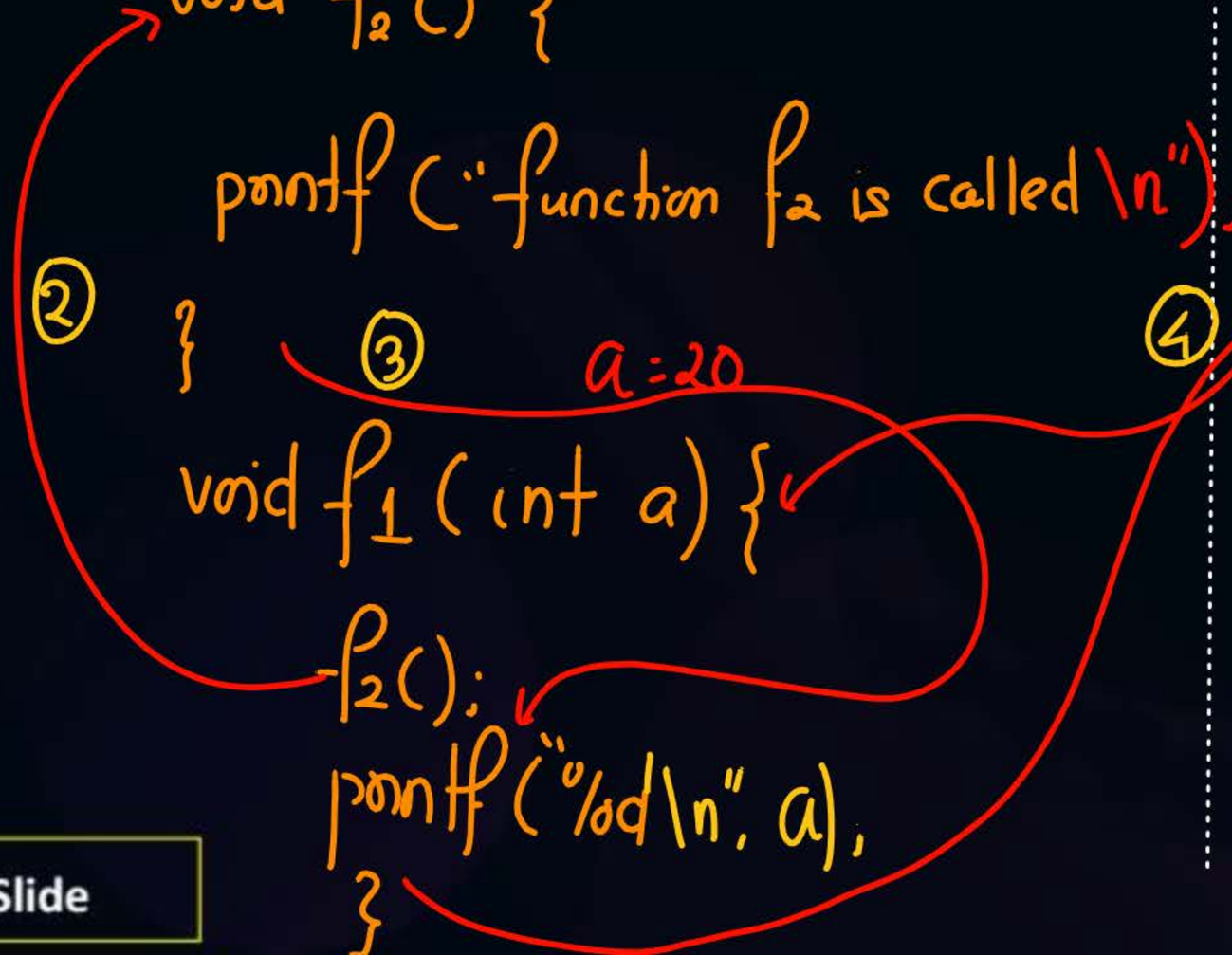


- Topic — Multiple function
- Topic — Recursion (Mathematical aspect)
- Topic
- Topic
- Topic

Slide

Function

```
#include<stdio.h>
void f2() {
    printf("function f2 is called\n");
}
void f1(int a) {
    f2();
    printf("%d\n", a);
}
int main(){
    int i=20;
    f1(i); //
    f2();
    return 0;
}
```




```
function f2 is called
20
function f2 is called
```

Slide



Function

Nested function : within a function if another function defined then we call it as Nested function.



Recursion

Recursion is a problem Solving Technique in which Solution of problem expressed as smaller instance of Same problem.



Recursion

In program Recursion is defind as a function which call itself

function calling itself

Slide



Recursion

1. Base Condition :- Are those condition which terminates the Recursion without Recursive function call.

2. progress :- The next Recursive function call should be on updated value otherwise it leads to infinite function call.

Slide



Recursion

factorial

$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

$0! = 1 \qquad 1! = 1$

$$n! = \lfloor n = n(n-1)(n-2)(n-3) \times \cdots \times 1$$

$$\lfloor (n-1) = (n-1)(n-2)(n-3) \times \cdots \times 1$$

$$\lfloor n = n \times \lfloor (n-1)$$

Slide



Recursion



$$\underline{|n} = \begin{cases} n \times \underline{|n-1} & n \geq 1 \\ 1 & n = 1 \end{cases}$$

Progress (Smaller value)

Base condition

Mathematical formulation

Slide



Recursion

```
#include<stdio.h>
int fact(int n){
    if(n==0||n==1)
        return 1;
    else
        return n*fact(n-1);
}
```

```
int main(){
    int a=5;
    printf("%d", fact(5));
    return 0;
}
```

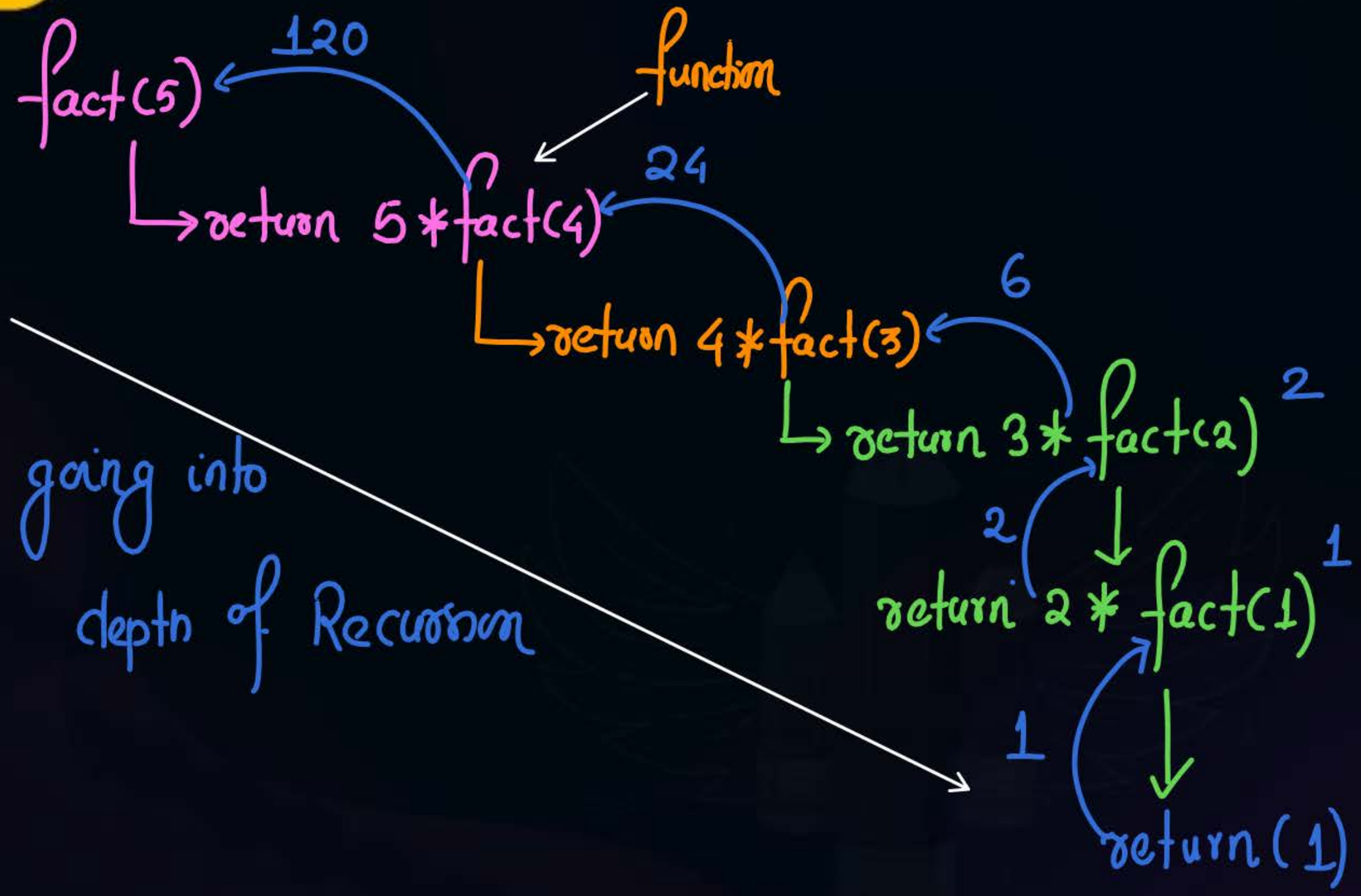
// 120 print

Slide



Recursion

Recursion Tree

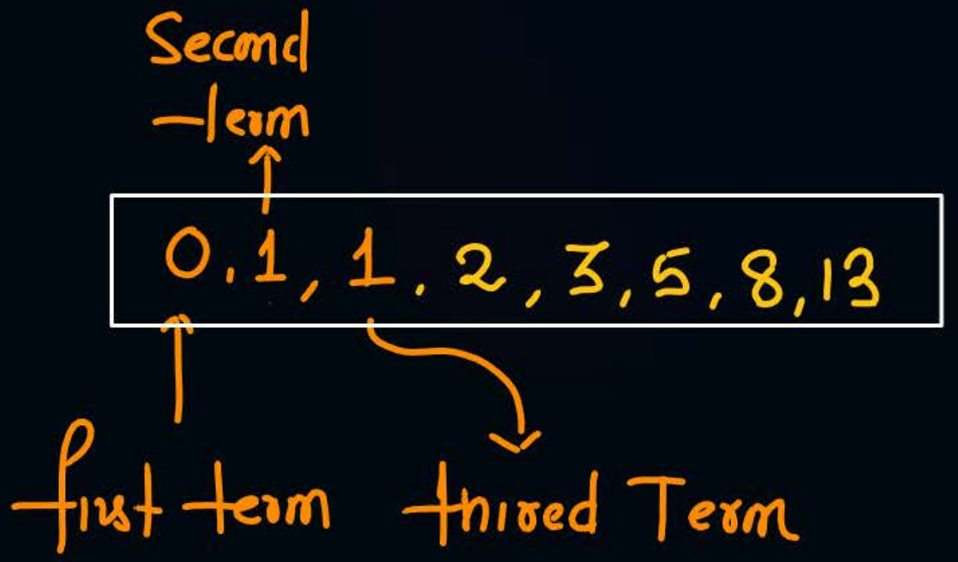


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Recursion

2. Fibonacci Series



Next term is Summation of previous two terms

Slide



Recursion

If we define a function fib(n) which is n^{th} term in fibonacci Sequence then fib(n)

$$fib(n) = \begin{cases} fib(n-1) + fib(n-2), & n > 1 \\ 1 & — \quad n = 1 \\ 0 & — \quad n = 0 \end{cases}$$

Slide



Recursion

if fib(n) is n^{th} term

$(n-1)^{th}$ term will be = fib(n-1)

$(n-2)^{th}$ term will be = fib(n-2)

Slide



Recursion

```
#include<stdio.h>
int fib(int n) {
}
```

```
int main() {
  int a=10,
  printf("%d", fib(10));
  return 0,
}
```



2 mins Summary

Topic Multiple function call

Topic Recursion

Topic factorial function

Topic fibonacci Series

Topic

THANK - YOU

